COURSE NOTES: ADVANCED ARTIFICIAL INTELLIGENCE
MODULE 4: RETRIEVAL-AUGMENTED GENERATION (RAG)

1. INTRODUCTION TO RAG
Retrieval-Augmented Generation (RAG) is an architectural framework designed to mitigate hallucinations in Larg

2. CORE THREE-STAGE PIPELINE
A standard RAG architecture operates across three sequential phases:
- Ingestion Phase: Raw unstructured documents (e.g., lecture PDFs, research papers) are parsed, sanitized, spli
- Retrieval Phase: When a student submits a natural language query, the system vectorizes the query using the
- Generation Phase: The retrieved top-k passages are combined with a strict grounding prompt and fed into the g

MODULE 4: VECTOR SIMILARITY & DENSE RETRIEVAL

3. DENSE RETRIEVAL VS SPARSE RETRIEVAL

Traditional search systems rely on sparse lexical matching like BM25 or TF-IDF, which match exact keyword tok

4. MATHEMATICAL FORMULATION: COSINE SIMILARITY

To measure the semantic alignment between a query vector q and a document chunk vector d, cosine similarity

CosineSimilarity(q, d) = (q . d) / (||q|| * ||d||)

When vectors are L2-normalized such that ||q|| = 1 and ||d|| = 1, the cosine similarity simplifies directly to the inne

CosineSimilarity(q, d) = q . d

5. EVALUATION METRICS FOR RAG SYSTEMS

- Hit Rate at K (Hit@K): The fraction of queries where at least one ground-truth relevant passage appears in the
- Mean Reciprocal Rank (MRR): Evaluates ranking precision based on the reciprocal rank of the first relevant do

MRR = (1 / |Q|) * Sum_{i=1}^{|Q|} (1 / rank_i)

MODULE 4: CHUNKING STRATEGIES & FAISS INDEXING

6. CHUNKING PARAMETERS AND TRADE-OFFS

Chunking is the process of partitioning lengthy documents into manageable segments.

- Chunk Size: Defines the character or token length of each passage (typically 500 to 1000 characters). Oversize
- Chunk Overlap: The sliding overlap between consecutive chunks (typically 100 to 200 characters). Overlap ens

7. FAISS VECTOR INDEX ARCHITECTURE

Facebook AI Similarity Search (FAISS) is an open-source library optimized for high-performance similarity searc

- IndexFlatIP: Performs exhaustive, exact inner-product similarity search. When paired with normalized vectors, i
- Inverted File Index (IndexIVFFlat): Employs Voronoi cell clustering to partition the vector space for approximate

8. DEFINITION GLOSSARY:

- Parametric Memory: Knowledge stored implicitly within the neural network weights learned during pre-training.
- Non-Parametric Memory: External structured or unstructured documents indexed in a vector store for real-time
- Grounding: Restricting model responses exclusively to factual assertions verifiable in the retrieved context.